Takudzidza by Taku

Phase 0: product spec and build plan

For Takunda Maswi. Prepared 29 October 2026. Nothing has been built yet. This document asks for your approval and eleven decisions (page “Decisions for you”). The full technical version for agents is SPEC.md in the repo.

What I am asking

• Approve the direction: tools first, with Chris as the one tutor voice inside every tool.

• Answer decisions D1 to D11, or reply “go with your recommendations”.

• Send your test making skill file. Quizzes and mock exams wait for it.

• Set up the Convex project and deploy key, the Vercel project and the OpenRouter key as environment variables on this cloud environment. Until then I build in mock mode.

Credit window: this cloud session ends 5 November 2026 at 09:59 Harare time. If you approve today, the target is the foundation and notebooks fully working and tested, with study outputs started. I will not promise more than what passes its tests.

1. What Takudzidza is

A study platform for anyone from Form 1 upward: ZIMSEC O Level and A Level and Cambridge learners, university and college students, and adults studying for professional exams. It launches in Zimbabwe and is built so exam boards, languages and currencies are settings, not rewrites. Amai Learn hands children over at Form 1. Takubasa stays the adult work platform.

What changes from the July design

July planned an offline tutor running small models on the phone, with lesson packs written by teachers. This plan puts the learner’s own material at the centre, like NotebookLM, in a web app that installs on Android and downloads notebooks for offline use. Kept from July: the teaching rules live in code, the maths engine holds the answer, level controls, no past paper sharing, and cheap phones as a hard limit. Phone models and the school hub are paused, not dropped. The July documents move to an archive folder.

2. Chris, the tutor

Recommendation: the app is a set of study tools, and Chris is the single tutor voice inside each one. Not a mascot, not a companion, no character picture unless you choose one.

Why hints first: in a trial with about 1,000 high school maths students (Bastani and others, PNAS, 2025), students given plain GPT 4 scored 48 percent better in practice but 17 percent lower than students with no AI once it was taken away. A version built to give hints instead of answers largely removed that harm. I checked the paper’s details through RePEc, SSRN and Wharton’s write up; the PNAS site itself was blocked from this session. Note the wording: “largely removed”, not “did no harm”.

The rule in the app: every notebook is marked Studying or Homework and tests. For homework and tests Chris gives hints and questions, never the final answer or a finished essay. For studying Chris can explain fully. Maths always goes through the maths engine, which keeps the answer away from the AI.

Chris modes, each built like a Takubasa agent so new ones can be added the same way:

Mode	What it does
Summarise	Summary or study guide from your sources, each point linked to its page
Explain it simply	Plain explanation at your level
Quiz me	Quiz or mock exam at your chosen difficulty, with a marking key
Flashcards	Cards from your sources with Got it and Try again
Exam coach	Syllabus aligned practice, marked against a key
Essay feedback	Comments and questions on your draft; never rewrites work you will hand in
Audio overview	A spoken overview of your sources, plus read aloud
Study planner	A plan from your exam dates, subjects and hours per week

3. The core product

• Dashboard first. Exam dates, subjects and hours per week; a study plan; progress without streaks or guilt; weak topics from your quiz and card results.

• Notebooks. Add PDFs, Word files, phone photos of notes and textbook pages, pasted text and web links. Every answer links to the page it came from. Choose “Only my material” or “My material plus Chris’s ideas”; every addition from Chris is marked.

• Stronger citation check than Amai Learn. Amai Learn checks that a cited page belongs to the topic. Takudzidza also checks that a short quote really appears on that page, and drops or marks anything that fails.

• Study outputs. Summaries, study guides, flashcards with spaced repetition, quizzes and mock exams with marking keys, mind maps, audio overviews, read aloud and print packs.

• Exam focus. Practice aligned to ZIMSEC and Cambridge syllabus topics. Past paper practice only from papers you upload yourself; papers are never shared.

• Maths and science. I found your earlier maths tutor in taku dzidza local: a SymPy solver that solves, factors, simplifies and expands, with a simple router. Recommendation: rebuild that logic in TypeScript so it runs on the server and offline on the phone, and keep its cases as tests.

• Sharing. A teacher or lecturer shares a notebook with a class as view only. Study groups come later.

• Access. English first, ChiShona next, offline downloads of notebooks and audio, a low data mode, and a speed budget tested on a low end Android profile.

• Safety. Under 18s need a guardian’s consent before a profile exists (Cyber and Data Protection Act, section 26 according to DLA Piper’s summary; to be checked against the gazetted text). Amai Learn’s safety layers, retuned so normal school topics pass. Childline 116 shown to under 18s in distress. No companion behaviour, no ads, no training on learner data, zero data retention AI routes only.

4. Look and feel

Warm cream and butter, black pill buttons, large rounded cards, generous spacing, and the Lexend typeface (made for easy reading). The colours sit right beside Amai Learn’s cream, gold, coral and leaf, so the two products look related. I added one colour, Ink Soft, for secondary text.



Cream	Butter	Honey	Ink	Ink Soft	Msasa Red	Sage
#FFF8EC	#F8E2A8	#F2B33D	#111111	#5C5346	#B4452F	#4F6B54

Contrast check (WCAG AA needs 4.5 for normal text, 3 for large text):

Text colour	On Cream	On Butter	On Honey
Ink	17.89 pass	14.78 pass	10.15 pass
Ink Soft	7.15 pass	5.91 pass	4.06 large text only
Msasa Red	5.19 pass	4.29 large text only	2.95 fail
Sage	5.58 pass	4.61 pass	3.17 large text only
Honey	1.76 fail	1.46 fail	

Rules that follow: Honey is a fill only, never text. Only Ink sits on Honey. Msasa Red body text never sits on Butter. Cream text on Ink, Msasa Red or Sage passes. A test will enforce these.

Photography: realistic, modern learners with quiet African warmth, no rural scenes, no cartoons, no generated images in the app. Codex batch 1 (six photos) is in the repo. I wrote the style rules and ten new prompts in the repo (art/STYLE.md and art/ASSETS.md) for you to run on your PC. Until each photo lands, the app shows a labelled placeholder.

5. What we reuse

Need	Taken from
Reading PDFs, Word files and photos; source modes; citation check	Amai Learn
AI router with zero data retention and strict structured answers	Amai Learn
Safety filter, model checks, red team script	Amai Learn
Consent records, account delete, retention clean up	Amai Learn, Takubasa learner consent record
Read aloud, print packs, offline downloads	Amai Learn
Agent package format for Chris modes	Takubasa
Study agent tests (past paper refusal, syllabus honesty), humaniser rules	Takubasa
Number checking against sources	Takubasa
Budgets, usage, brains console, sign offs, admin console	Takubasa (later phase)
Maths solver	taku dzidza local (rebuilt in TypeScript)
Not reused now	TAKU dzidza v2 desktop app and phone model, Takubasa image brains and Studio

6. Competitors and local options

Prices come mostly from third party listings; the official Knowt and Khanmigo pages were blocked from this session, so treat them as indicative.

Product	Strength	Price seen	Our gap to fill
NotebookLM	Own sources, cited answers, audio overviews, cards and quizzes	Free tier; paid via Google AI plans, figures conflict	No exam focus, no hint mode, no guardian consent, no EcoCash
Quizlet	Flashcards, Learn mode	About USD 7.99 a month or 35.99 a year	AI modes paywalled, not syllabus aware
Knowt	Free notes to flashcards	Listings conflict	Not cited to page, not local
Studocu	Shared course documents, AI summaries	Varies by country	Built on shared documents, which we will not do
Khanmigo	Tutor that holds back answers	About USD 4 a month for learners	US curriculum, not your own sources
Skoolie Zimbabwe	ZIMSEC and Cambridge video lessons	Not seen	Video only
Dzidzo paDen	Free ZIMSEC papers and notes on WhatsApp	Free	No tutor or tools
Wagona Maths	ZIMSEC and IGCSE maths	Not seen	Maths only

7. Build plan

Every phase must pass lint, type checks, unit tests, the production build and browser tests, all with the AI mocked. Live AI calls only run with a spending cap I state first. One commit per phase.

Phase	What gets built	Gate
0	This plan, the PDF, art rules and prompts	Your approval and decisions
1 Foundation	App shell from Amai Learn, colours and type, accounts, guardian consent, delete account, safety filter, mocked AI	All checks green, consent tested
2 Notebooks	Add sources, ask Chris, page citations with quote check, source mode toggle	No uncited claim in Only my material
3 Study outputs	Chris modes, summaries, cards with spaced repetition, quizzes and mocks, mind maps, read aloud	Red team pass; you review 10 tests
4 Dashboard	Goals, study plan, progress, weak topics	Onboarding to first plan tested
5 Maths and exams	Maths engine, hint ladder, syllabus practice, uploaded past papers	No answer leaks in red team
6 Reach	Class sharing, offline downloads, low data mode, ChiShona, audio overviews	Offline and low end phone tests
7 Business	Plans, EcoCash payments, school licences, admin and budgets	Capped live payment test; you sign

8. Pricing proposal

What I could verify: Khanmigo charges learners about USD 4 a month; Quizlet Plus about USD 35.99 a year; Econet has advertised 1 GB of data for USD 1 (offer date to recheck), and Paynow lists a 30 day private WiFi bundle of 4 GB plus 1 GB off peak for USD 9. I found no public price for a Zimbabwean revision app paid by EcoCash. The AI cost per learner is unknown until we measure it in Phase 2.

Proposal to test with about 20 learner interviews before any billing code:

• Free: 3 notebooks, a daily allowance of Chris, unlimited flashcards, 1 offline notebook.

• Student: USD 2 a month or USD 5 a term by EcoCash. Term pricing matches how families budget school costs.

• Schools and universities: per learner per year, quoted per institution; teacher accounts free.

• Zero rating: approach Econet and NetOne once a pilot shows real use.

9. Decisions for you

#	Decision	My recommendation
D1	Chris as tutor voice inside tools, or tools only	Tutor voice inside tools
D2	Pricing	As in section 8, after interviews
D3	First exam boards	ZIMSEC O Level and A Level, plus Cambridge IGCSE topic lists
D4	Launch subjects for syllabus practice	Mathematics, Combined Science, English Language; every subject works through notebooks
D5	Palette and Lexend	Approve, with Ink Soft added
D6	Maths engine	Rebuild your SymPy solver in TypeScript
D7	Visible Chris character	None
D8	Search inside notebooks	Use a zero data retention embeddings route, checked on the day of building
D9	Platform	Installable web app first, Android wrapper later, phone models paused
D10	Your test making skill file	Please send it
D11	Adult distress helpline	You name one; until then a general line to seek a doctor or emergency services

10. Risks

• Quote checks on scanned pages may miss due to reading errors; we will tune the match on real test files.

• AI cost per learner is not known yet; spending caps and a cheaper model fallback limit exposure.

• I read summaries of the data protection law, not the full gazetted text; it must be checked before launch.

• Learners may mark homework as studying; maths hints are enforced by the engine regardless.

Sources

Checked 29 October 2026. Full list with links is in SPEC.md section 13 and docs/SOURCES.md.

Bastani and others, Generative AI without guardrails can harm learning, PNAS 122(26), 2025. https://econpapers.repec.org/RePEc:nas:journl:v:122:y:2025:p:e2422633122 and https://knowledge.wharton.upenn.edu/article/without-guardrails-generative-ai-can-harm-education/

NotebookLM limits: https://freeacademy.ai/blog/notebooklm-free-vs-plus-2026

Quizlet: https://nibble-app.com/blog/quizlet-cost · Khanmigo: https://toolradar.com/tools/khanmigo/pricing · Knowt: https://toolradar.com/tools/knowt/pricing

Studocu: https://apps.apple.com/app/id1460235511 · Skoolie Zimbabwe: https://mwm.ai/apps/skoolie-zimbabwe/6782972508

Econet data: https://techzim.co.zw/econet-now-offering-1-gb-for-1-in-latest-data-bundle-adjustment and https://paynow.co.zw/blog/econet-private-wifi-bundles-cost-validity-and-how-to-buy-online/

Children’s data: https://www.dlapiperdataprotection.com/index.html?c=zw&t;=law · Childline 116: https://www.unicef.org/zimbabwe/stories/zimbabwe-free-helpline-116-spells-hope-children